



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

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In words

Circle your Teacher's Name: Mr Bartoshefski Mrs Fraser-Jones Ms Horley Dr Kang
Ms Lastari Ms Nathan Mr Singh

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	8	
9	9	
10	10	
11	12	
12	10	
13	11	
14	10	
15	10	
16	10	
17	8	
Section 2 Total	98	
Weighted ($\times 0.6633$)	65%	

Section Two: Calculator-assumed**65% (98 Marks)**

This section has **ten** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

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Question 8**(8 marks)**

Jian wants to transfer 1100 Australian dollars to his sister who lives in Singapore and will carry out the transaction using his own bank (ANZ), PayPal or Wise. The table below shows the exchange rate (how many Singapore dollars that one Australian dollar will buy) and the transfer fee charged by each company in Australian dollars. Jian will pay the transfer fee upfront to ensure that his sister receives the full amount of the transfer in Singapore dollars.

	Own bank (ANZ)	PayPal	Wise
Exchange rate	0.8464	0.8366	0.8750
Transfer fee	\$9.00	5% of transfer*	\$0.83 + 0.55% of transfer

* A minimum transfer fee of \$0.99 and maximum transfer fee of \$5.99 applies for PayPal.

(a) Which company will charge Jian the least transfer fee? Justify your answer. (3 marks)

(b) Determine the amount, in Singapore dollars, his sister would receive if Jian used the most favourable exchange rate offered. (2 marks)

(c) Jian's sister has told him that 1100 Australian dollars is not enough because she needs 1400 Singapore dollars. Determine the total funds, in Australian dollars, that Jian must have to afford this transfer using Wise, including the transfer fee. (3 marks)

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Question 9**(9 marks)**

- (a) A supermarket customer assistant was employed at a base rate of \$22.40 per hour. Calculate his total earnings in a week when he worked for 36 hours at his base rate and for 5 hours 30 minutes of over-time, for which he was paid time-and-a-half. (3 marks)
- (b) Another person who ran an instore promotion for air fryers in the supermarket was paid a weekly retainer of \$245 plus 14.5% commission on the total value of the air fryers that the supermarket sold. Calculate the earnings for this person in a week when 26 small and 16 large air fryers were sold for \$119 and \$229 respectively. (3 marks)
- (c) The supermarket sells the same yoghurt in two different sized cartons: 125 g for \$2.40 and 420 g for \$8.19. Use the unit cost method to compare the prices of each size carton and hence state which size represents the best value. (3 marks)

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Question 10**(10 marks)**

Giovanna made some gelato and left it to cool in a cylindrical glass container with an internal diameter of 17 cm.

- (a) The container is filled with gelato to a uniform depth of 14 cm. Determine the volume of gelato that Giovanna has made, to the nearest cm^3 . (3 mark)

Previously, Giovanna had made 2650 cubic centimetres of gelato and used it to fill right circular waffle cones with internal radius 2.5 cm and height 11 cm as treats for his friends.

- (b) Determine the volume of gelato required to fill one of these cones level with its circular top. (2 marks)

- (c) How many complete cones could Giovanna fill? (1 mark)

(d) What volume of gelato was left over?

(1 mark)

Giovanna used the leftover gelato to make two types of small frozen shapes - cubes with a side length of 14 mm and spheres with a radius of 9 mm.

(e) Which shape used the least amount of gelato, and how much less gelato than the other shape? (3 marks)

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Question 11**(12 marks)**

Tamsin has prepared a weekly budget to check that she can afford to run a second-hand petrol fuelled car using the income from her casual job and government allowance.

Budget item	Weekly amount (\$)
Loan repayment	64
Registration and compulsory insurance	29
Comprehensive insurance	x
Petrol	45
Servicing	11
Roadside emergency assistance	3

- (a) Name a budget item from the table above that is an example of
- (i) a fixed expense. (1 mark)
- (ii) a discretionary expense. (1 mark)
- (b) Assuming there are exactly 52 weeks in a year, determine
- (i) the annual amount that Tamsin has budgeted for petrol. (1 mark)
- (ii) the value of x in the table (the weekly amount Tamsin has budgeted for comprehensive insurance) if her insurance policy costs \$78 per month. (2 marks)

Tamsin is eligible for a maximum fortnightly Youth Allowance payment of \$425, but the actual amount she receives is reduced because she has a casual job from which she earns \$360 per week.

Youth Allowance payments reduce according to the following rules:

Amount your payment reduces by if you earn between \$480 and \$575 per fortnight	Amount your payment reduces by if you earn more than \$575 per fortnight
50 cents for each dollar over \$480	\$47.50 plus 60 cents for each dollar over \$575

- (c) Determine the Youth Allowance payment made to Tamsin each fortnight. (3 marks)

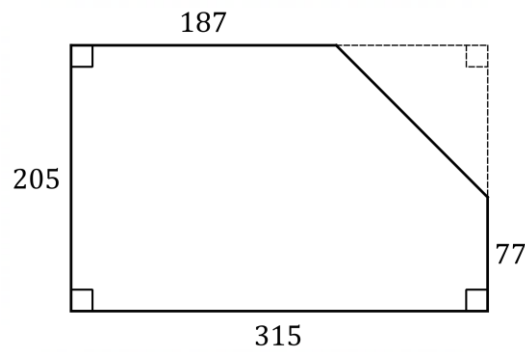
Tamsin cannot afford to spend more than 35% of her total earnings and allowance to run a car.

- (d) Will she have to find a way to reduce her budget? Justify your answer. (4 marks)

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Question 12**(10 marks)**

A manufacturer sells a range of rectangular timber-framed windows with a clipped corner. Four of the edge lengths of a piece of glass used to glaze one of these windows are shown in the diagram below, where the dashed line shows the triangular corner that is removed. The lengths are in centimetres and the diagram is not drawn to scale.



- (a) Determine the area of glass removed when the triangular corner is clipped. (3 marks)

- (b) Determine the area of glass that remains once the corner has been clipped. (2 marks)

- (c) Show use of Pythagoras' Theorem to determine the length of cut required to clip the corner. (2 marks)

Just before it is assembled in the timber frame, a thin layer of sealing tape is applied to the perimeter of the clipped pane of glass. The sealing tape costs 35 cents per metre.

- (d) Determine the cost of the sealing tape applied to the clipped pane of glass. (3 marks)

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Question 13**(11 marks)**

A financial advisor rents a small office to work in and meet clients.

- (a) A year ago, the annual rent payable for the office was set at \$460 per square metre. This amount is subject to an annual increase of 2.5% plus the percentage increase in the consumer price index over the previous year.

The consumer price index rose from 124.2 to 132.4 over the past year.

- (i) Show that the percentage increase in the consumer price index was 6.60%.
(1 mark)

- (ii) Determine the new annual rent payable for the office given that it has a floor area of 62.5 square metres.
(3 marks)

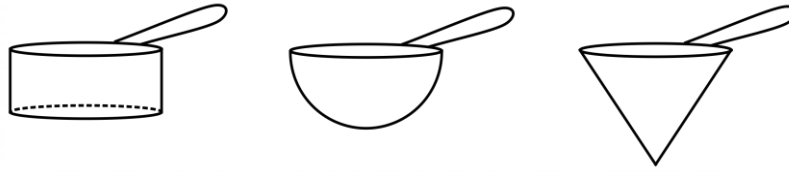
- (b) A client of the financial advisor wanted a 60 day loan of \$1095. If the loan attracted simple interest at 12.5% per annum, determine the total amount that the client would have to repay at the end of the loan period.
(3 marks)

- (c) Another client was advised to invest a sum of \$22 500 for 18 months in a product that paid interest at a rate of 3.9% per annum compounded quarterly. Determine the total interest this investment would earn. (4 marks)

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Question 14**(10 marks)**

A student has designed three metal scoops with identical wooden handles that can all be used by a baker to scoop exactly 150 grams of flour out of a large bin of the ingredient. They are shown in the diagram below, not drawn to scale.



One scoop is cylindrical with a radius of 4.5 cm and a height of 3 cm, one is hemispherical with a radius of 4.5 cm, and the other is conical with a radius of 4.5 cm and a perpendicular height of 9 cm.

- (a) Determine the area of metal required to make the cylindrical scoop. (2 marks)
- (b) Determine the area of metal used to make the hemispherical scoop. (2 marks)
- (c) Determine the slant height of the conical scoop and hence state the area of metal required to make the conical scoop. (3 marks)

- (d) By first calculating the volume of any one of the scoops, determine the volume of one gram of flour. (3 marks)

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Question 15**(10 marks)**

Five remote climate monitoring stations are numbered from 1 to 5. Due to their locations, not all stations can transmit data directly to the other stations. A list of all possible direct transmissions is given below.

- Station 1 can transmit data to Station 3
- Station 2 can transmit data to Station 3 and to Station 4
- Station 3 can transmit data to Station 2 and to Station 5
- Station 4 can transmit data to Station 1 and to Station 3
- Station 5 can transmit data to Station 4

Furthermore, no station can transmit data to itself, nor can it ever receive its own data back.

- (a) Construct matrix N to represent this transmission network, with stations in numerical order and with element $n_{i,j}$ equal to 1 if station i can transmit data to station j , and equal to 0 if not. (2 marks)

- (b) Determine matrix M , where $M = N^2$. (2 marks)

- (c) State the value of element $m_{2,2}$ in matrix M and explain why it contradicts one of the conditions in the introduction to this question. (2 marks)

- (d) Replace all the ones on the leading diagonal of matrix M with zeros, to form matrix P .
(1 mark)

- (e) Determine matrix $Q = N + P$. (1 mark)

- (f) Use matrix Q to state, with justification, which station(s) can transmit data to all other stations either directly or via one other station. (2 marks)

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Question 16**(10 marks)**

Danielle's bedroom is a rectangular room that measures 4.65 m by 6.35 m and she has created a scale drawing of it using a scale of 1: 25.

- (a) Would her drawing fit on a 20 cm by 25 cm sheet of paper? Justify your answer. (3 marks)
- (b) On the floor of her bedroom she has a circular rug with a radius of 1.05 m. It is split into different coloured sectors, the green one having a central angle of 55° . Determine
- (i) the area of the green sector of the actual rug. (2 marks)
- (ii) the area, in square centimetres, of the green sector on her scale drawing of the rug. (2 marks)
- (c) A cube of foam in her bedroom that her friends sit on needs a new cover. On her scale drawing, the length of one side of the cube is 22 mm. Allowing an extra 20% for seams and wastage, how many square centimetres of fabric should be bought to make a new cover for the cube? (3 marks)

Question 17

(8 marks)

A borrower must make n monthly repayments of A dollars to repay a loan of L dollars when the interest rate on their loan is $R\%$ per annum compounded monthly. To calculate the monthly repayment A , the following formulas are used:

$$A = \frac{Lr}{b}, \text{ where } r = \frac{R}{1200} \text{ and } b = 1 - (1 + r)^{-n}.$$

- (a) Lars has negotiated a loan of \$52 000 to buy a new EV. The interest rate on his loan is 7.8% per annum compounded monthly and he will make 60 monthly repayments.

(i) Calculate the value of r for this loan. (1 mark)

(ii) Calculate the value of b , rounding your answer to 6 decimal places. (2 marks)

(iii) Show that the monthly repayment Lars must make is \$1049.40. (1 mark)

- (b) Leigh has budgeted monthly repayments of no more than \$150 to buy an electric scooter priced at \$3250. She has been offered a loan for the full purchase price with 24 monthly repayments at an interest rate of 8.4% per annum compounded monthly. If Leigh goes ahead with the loan, will the repayments satisfy her budget? Justify your answer. (4 marks)

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Supplementary page

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